

Backtest -> Paper Test -> Live

What we did in the backtests and why it earned a paper test, what each of the five paper-test models is doing, and what happens after. Plain-English, real numbers.

Part 1 - The backtests: what we did, and why it was worth continuing

A backtest replays years of past market data and measures how a strategy would have done. The danger is 'data leakage' - accidentally letting the model peek at the future, which makes junk look brilliant. Eleven earlier versions did exactly that. So we rebuilt the whole thing to make cheating structurally impossible.

How we made it trustworthy

- Purged, embargoed walk-forward validation - the model is only ever judged on data it never trained on.
- Executable leakage tests that ABORT the run if any feature can see the future.
- A framework self-check that proves the engine recovers real signal, invents none on noise, and halts on leaks.
- An independent backtest cross-check (VectorBt + numpy) confirming the numbers are correct arithmetic, not a bug.

What we tried, and what happened

Attempt	Result
Versions 1-11 (early ML pickers)	Looked amazing - but it was data leakage. Fake. Discarded.
Version 12 (leak-proof rebuild)	Honest results, much smaller. The real turning point.
+ SEC fundamentals	A fair test PROVED it made predictions worse. Removed from the core.
+ SEC insider (Form 4)	Same - lowered real skill. Removed from the core.
+ complex models / stacking	No improvement over the simple model.
Final: minimal model	volatility + relative-ranking, regime-gated: the ONLY version with a real, statistically significant edge.

The validated result (real data, 2018-2026, after costs)

Our strategy vs the S&P 500 and cheap low-vol ETFs

Real data, out-of-sample 2018-2026, after costs. Green = best in each row.

Metric	Our Strategy	S&P 500	USMV	SPLV
Sharpe (return per unit of risk)	0.74	0.74	0.64	0.53
Sortino (per unit of downside risk)	0.81	0.91	0.76	0.62
Calmar (return vs worst drop)	0.47	0.39	0.28	0.21
Max drawdown (worst crash)	-19.0%	-33.7%	-33.1%	-36.3%
Annual volatility	12.7%	19.3%	15.6%	16.5%
CAGR (yearly return)	9.0%	13.2%	9.1%	7.7%

Verdict: ties the S&P on risk-adjusted return with about half the drawdown, and beats the low-vol ETFs — a defensive alternative, not a market-beater.

Why it was deemed worthy to move on to paper testing

- The signal is statistically REAL, not luck: rank-IC 0.055 with a fold t-stat of 2.04 across 32 separate

out-of-sample windows (above 2 = significant).

- It survives reality: realistic costs, Monte-Carlo stress, and fat-tail stress don't break it.
- Independent engines agree: VectorBt + numpy confirm the metrics are correct arithmetic, not an engine bug.
- It clears the real bar: it beats the off-the-shelf low-volatility ETFs (USMV, SPLV) it would compete with - so it is not just a redundant version of something you can already buy.
- The value is genuine but DEFENSIVE: it ties the S&P on risk-adjusted return with about half the crash (-19% vs -34%). It does NOT beat the S&P on raw return - and we say so plainly.
- The ONE thing a backtest cannot settle - survivorship bias (the test universe is today's survivors) and whether live trading matches the backtest - can only be resolved going FORWARD. That is exactly what a paper test does, and why we moved to one instead of going live or quitting.

Part 2 - The five paper-test models: what each is doing

A backtest is the past; a paper test is the FUTURE, traded with fake money in real time - the honest, survivorship-free way to see if the edge holds. Five strategy 'variants' race side by side, every weekday, fully automated, ZERO real money. Each makes real decisions and records its real next-period return. They exist to answer one question: does the heavier machinery actually beat the simple validated strategy?

#	Model	What it is / what it's doing	The question it answers
1	equity_validated	The validated low-vol strategy from Part 1: stocks, price/volume signals, a 2-state 'risk-on/off' regime gate. The CONTROL arm.	The bar everything else must beat.
2	equity_full_goal	Stocks with ALL the system rules: 6-state regime detection, multi-horizon signal blend, expected-value gate, correlation checks, Kelly-capped sizing, position caps.	Does the full rulebook beat the simple version, on stocks?
3	crypto_full_goal	The same full rulebook applied to crypto (BTC, ETH + large-cap alts), with crypto-specific caps (30% class, 12%/position).	Does the system add value in crypto?
4	full_system	Models 2 + 3 combined under the capital structure ($\leq 70\%$ stocks, $\leq 30\%$ crypto, rest cash) PLUS a learning loop that shifts weight toward whatever is performing.	Does the complete engine beat the baseline?
5	full_system_max	Model 4 PLUS every available SEC data source (fundamentals + insider) - the same data our backtest showed HURT, now tested FORWARD to confirm that finding.	Does the extra data help or hurt, live?

Rules that apply to ALL five (the safety layer)

- Cash is the default: a model holds cash when expected value is negative, the regime is dangerous, or confidence is too low. It is never forced to be invested.
- A hard risk governor runs on each: max-drawdown kill-switch, daily-loss stop, and a freeze after 3 losing periods
- all force the model to cash automatically.
- Every decision is fully transparent: asset, action (BUY/HOLD/SELL/NO-TRADE), expected-value score, risk rating, confidence, position size, reasoning, and the exact data sources used.
- It needs ~20 trading days before the scoreboard is meaningful, and ~90 days for a real verdict.

Part 3 - What we do after the paper test (~3-6 months)

After 90+ trading days of honest forward results, each model faces a 'promotion gate'. It can graduate from fake money to real ONLY if it clears every one of these:

Gate check	Plain meaning
Positive real Sharpe	It actually made risk-adjusted money going forward, after costs.
Beats the baseline	The fancy machinery must out-perform model 1 - or we simply keep model 1.
Beats USMV / SPLV	It must still beat the cheap ETFs you could buy instead.
Risk rules respected	No drop worse than the stop allowed; it went to cash in stress.

Then one of three honest outcomes

- It clears the gate -> graduate to LIVE paper, then a SMALL amount of real capital, scaling only as it keeps proving itself (never all at once).
- It only ties the baseline -> deploy just the simple validated strategy; drop the extra complexity.
- Nothing clears the gate -> the honest answer is 'no durable edge - just buy the ETF.' Concluding that truthfully is itself a success; it is what eleven earlier 'amazing' versions failed to admit.

If it graduates to live (the plan)

- Host: a small always-on cloud server (DigitalOcean droplet) runs the daily decision and submits orders to Alpaca. Alpaca is the broker, not the host.
- Must be built first: a gated live adapter, a full rebalancing order manager (sell/reduce, not just buy), and a one-command 'flatten to cash' kill-switch.
- Start tiny. Expect live to be a bit WORSE than paper (real slippage/fills). The edge is defensive - lower

drawdown, roughly matches the market - not a way to get rich fast.

Bottom line: we built a testing engine that refuses to lie to us, used it to find a small but real defensive edge that beats cheap alternatives, and we are now confirming it forward with fake money across five models before risking a cent. Nothing trades real money today.